Open-ended research with AI agents

Using today's most capable models on hard, open questions without losing control

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01 · THE PROBLEM

Smart models fail beautifully.

1 More and more research goes to capable models.

2 It fails silently: one hidden assumption sinks it.

3 More output than we can check: you're the bottleneck.

A gem in the haystack.

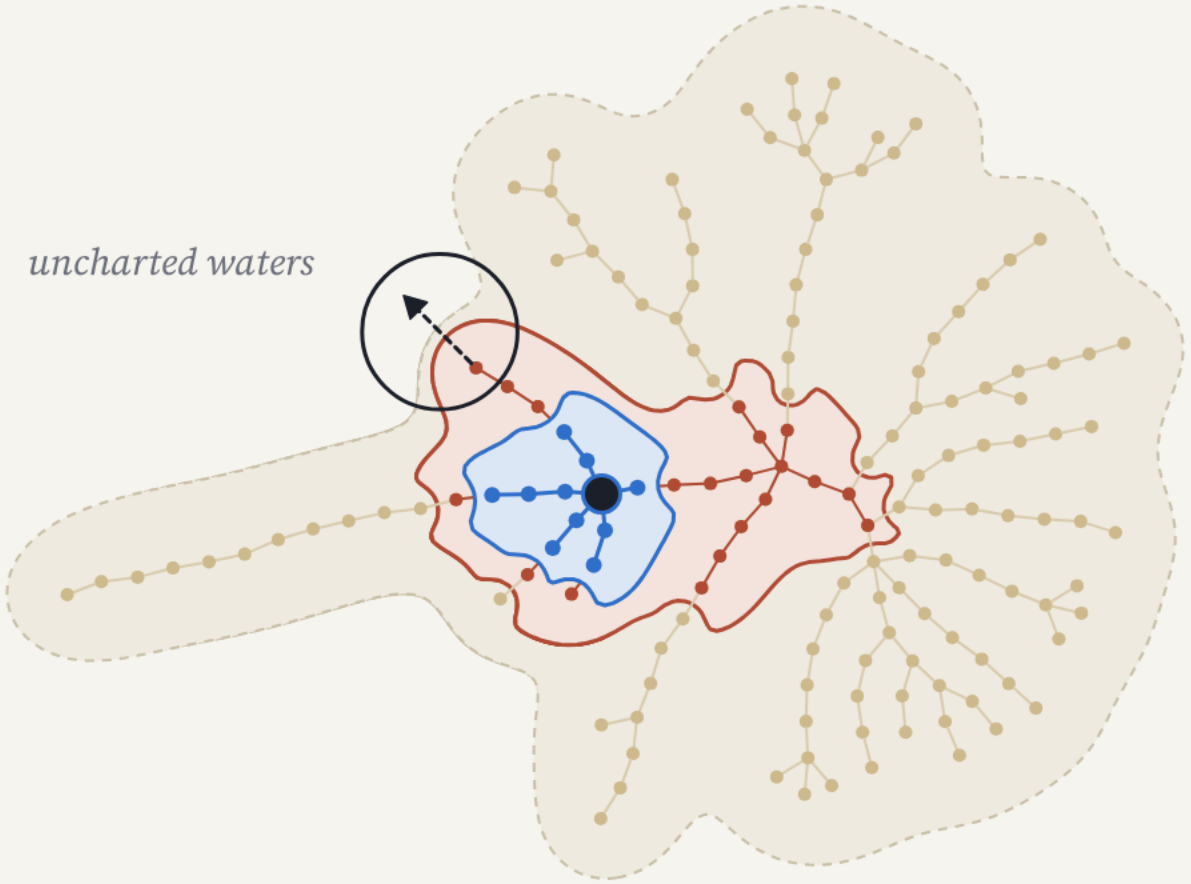




How do we find the gem,
and stay in control?

02 · THE STRATEGY

The strategy at a glance.



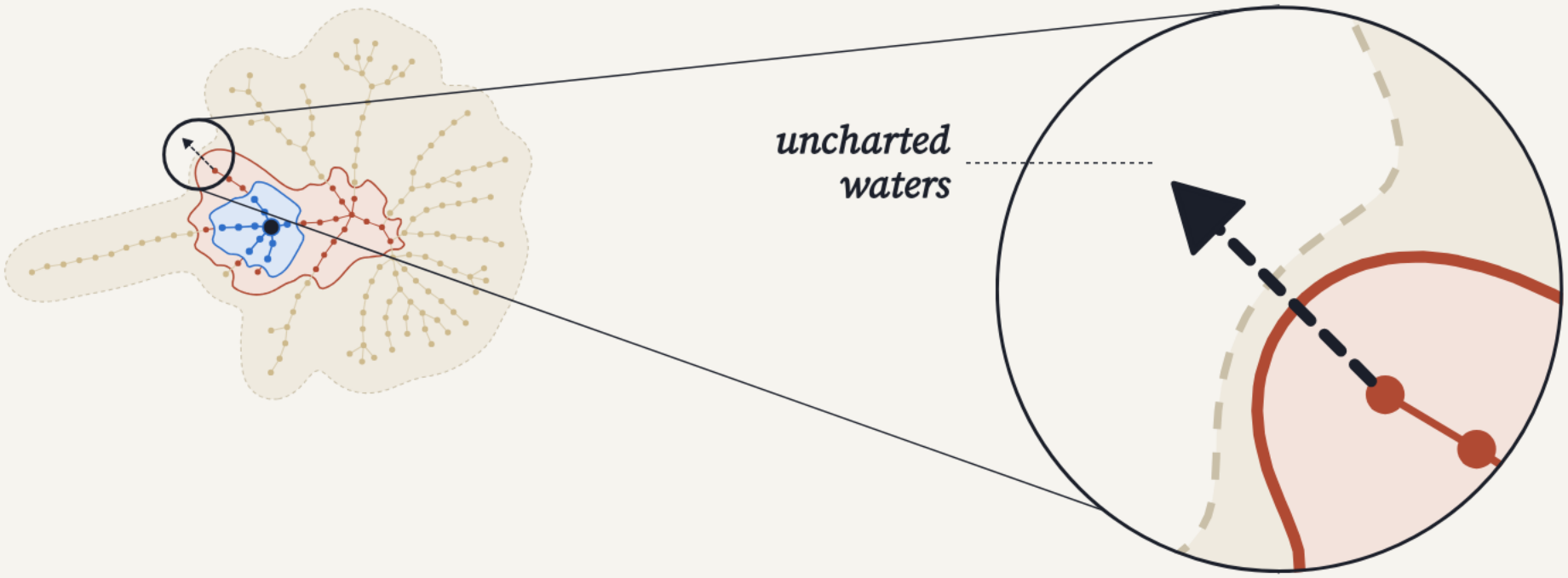
Open-ended question
where every thrust starts

Research thrusts
fresh exploration, pushed forward

Established core
audited and relevant

The gem: a paper
carved out, audited for novelty, significance

That's the gem in the haystack.



How do we make each push succeed?

03 · INSIDE ONE THRUST

The multi-agent engine behind every thrust.

ONE THREAD
Why one long thread fails

INDEPENDENCE
Fresh agents, different angles

OPPOSITION
An author and a critic

COST
A pyramid, and cheap checks

Many agents, well organized, make the push feasible.

03 · INSIDE ONE THRUST

The multi-agent engine behind every thrust.

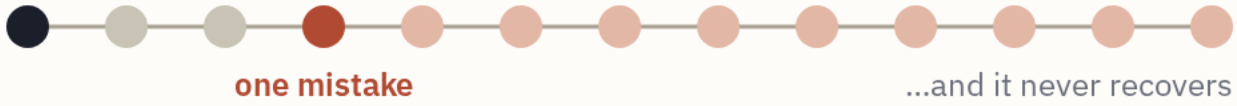
ONE THREAD
Why one long thread fails

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ONE LONG THREAD



· Every dead end piles into one context
· An early mistake anchors the rest
· “Be unbiased” can't undo it

03 · INSIDE ONE THRUST

The multi-agent engine behind every thrust.

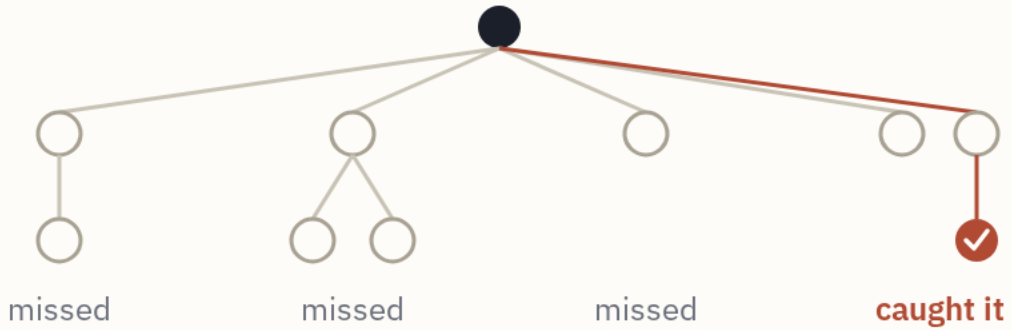
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INDEPENDENCE: FRESH AGENTS, DIFFERENT ANGLES



· Each starts clean, sees only its folder
· Several miss; one catch is enough
· Skills: parallel proof routes, gate reviews

03 · INSIDE ONE THRUST

The multi-agent engine behind every thrust.

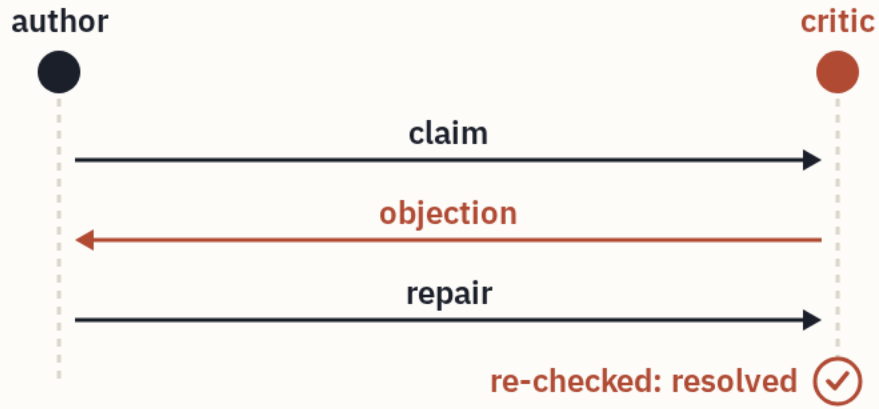
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OPPOSITION: AN AUTHOR AND A CRITIC



· Opposing roles, round after round
· Skills: the hostile audit in formulate-problem
· A fresh review after every fix

03 · INSIDE ONE THRUST

The multi-agent engine behind every thrust.

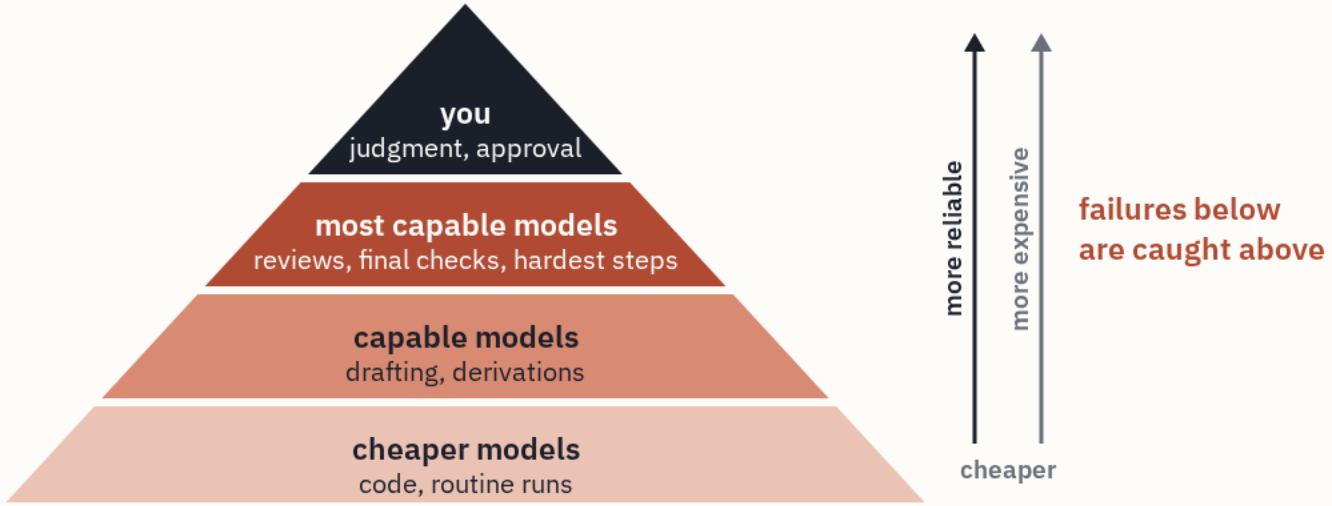
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COST: A PYRAMID, AND CHEAP CHECKS



If a claim can be tested, test it

A script refutes in seconds. Skills: small-case checks, counterexample search.

How do we do this in practice?

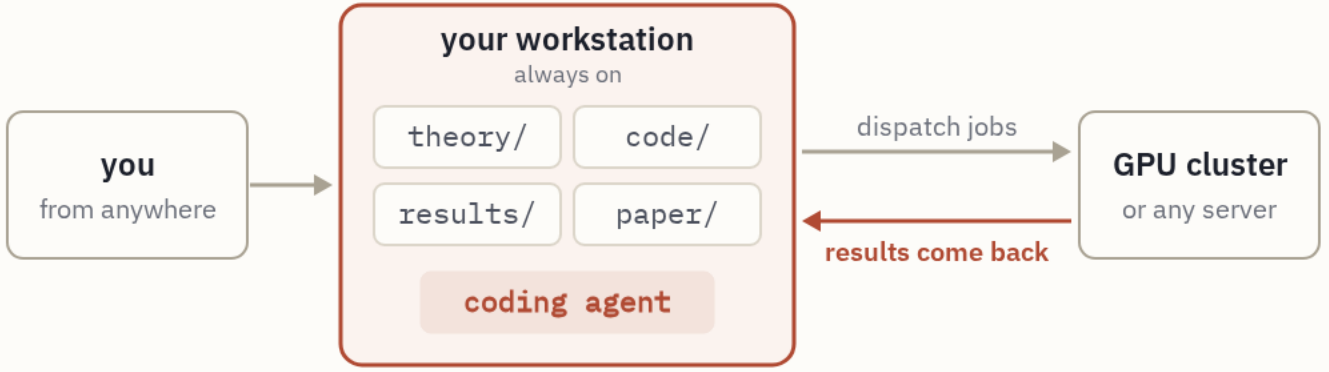
Three foundations that make it practical.

CONTEXT
A coding agent on one workstation

PERSISTENCE
Everything under version control

REUSE
Projects and skills

CONTEXT: EVERYTHING IN ONE PLACE



- A coding agent, not a chat
- Theory, code, results, paper: all in reach
- Works overnight, waits for runs

Three foundations that make it practical.

CONTEXT

A coding agent on one workstation

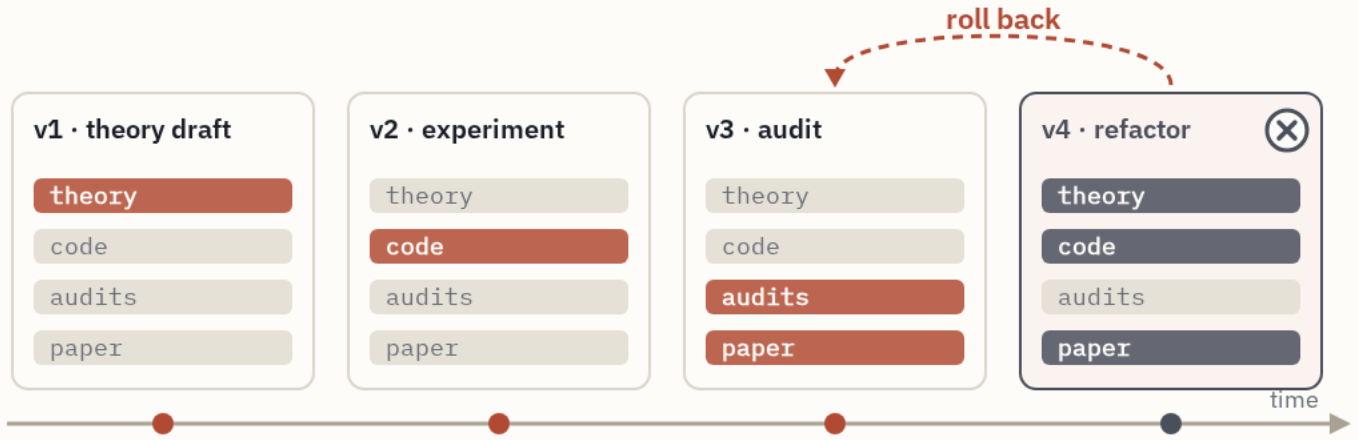
PERSISTENCE

Everything under version control

REUSE

Projects and skills

PERSISTENCE: EVERY SNAPSHOT HOLDS EVERYTHING



· Not just code: theory, audits, paper

· Roll back a bad agent or decision

· Big refactors become safe

Three foundations that make it practical.

CONTEXT

A coding agent on one workstation

PERSISTENCE

Everything under version control

REUSE

Projects and skills

REUSE: PROJECTS AND SKILLS



- A project: a folder with shared rules
- Skills are plain files: portable across models
- Everything in this repo is a skill

Everything from here is a skill.

05 · FIVE SKILLS

Five skills: from open-ended question to paper.

A whole project
one big, open question

ORGANIZE
`organize-research-project`
Many thrusts → one core

the core becomes a paper →

AUDIT
`review-paper`
Paper → rigorous audit

↺ findings start the next round

↓ every thrust runs these three

Each thrust
one targeted question

FORMULATE
`formulate-problem`
Open → precise question

→

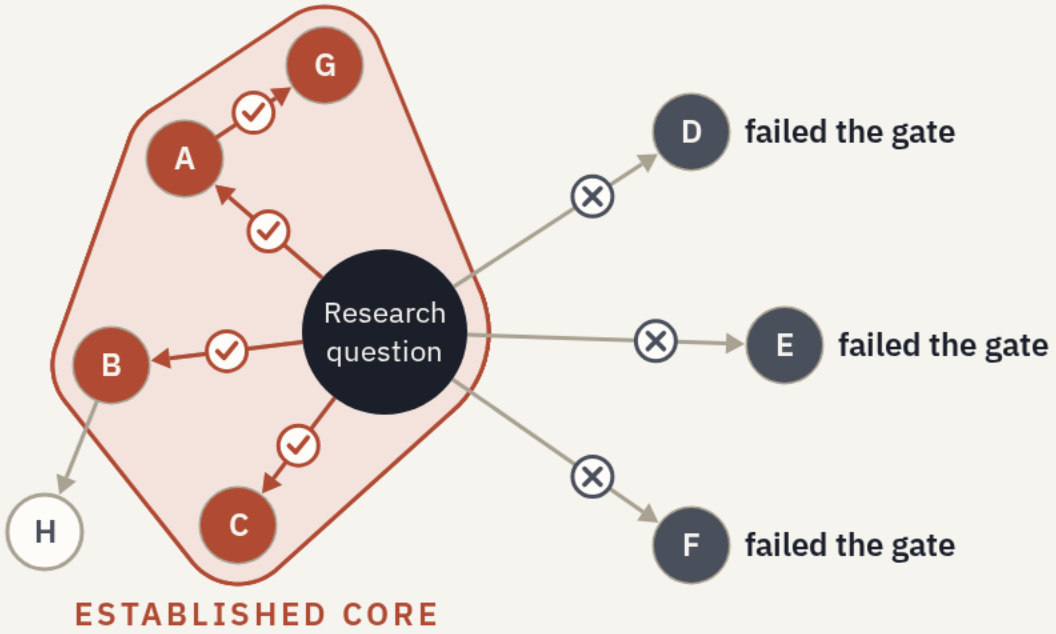
SOLVE
`solve-problem`
Question → checked proof

→

LEARN
`teach-material`
Result → understood

`organize-research-project`

Organize: isolated thrusts, a gated core.



Inside each study: formulate, solve, learn.

1 Isolated studies

Own question, inputs and folder; blind to the others by default.

→ **the widest exploration**

2 A promotion gate

Correct: an independent review
Coherent: notation, interfaces, tests
Relevant: your approval

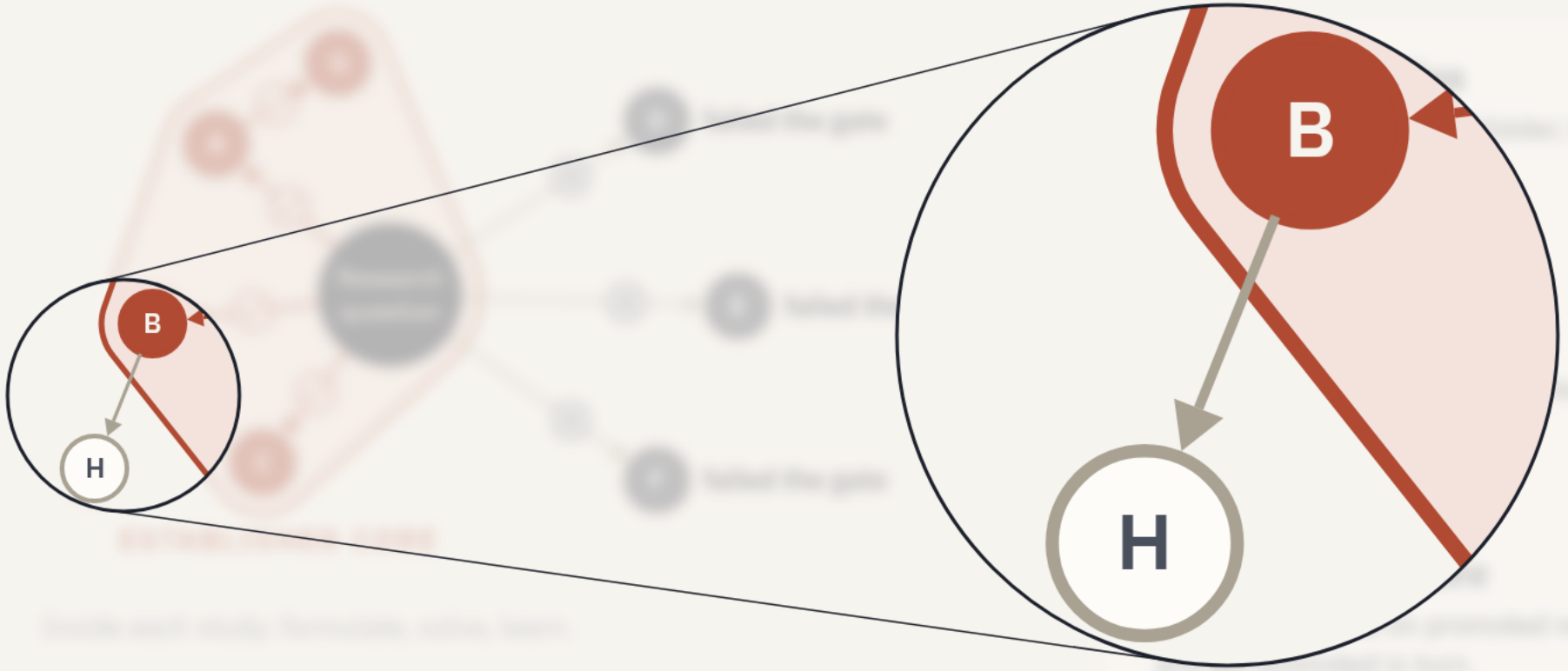
→ **high signal-to-noise**

3 Build on the core

New studies build on promoted results,
and get promoted in turn.

organize-research-project

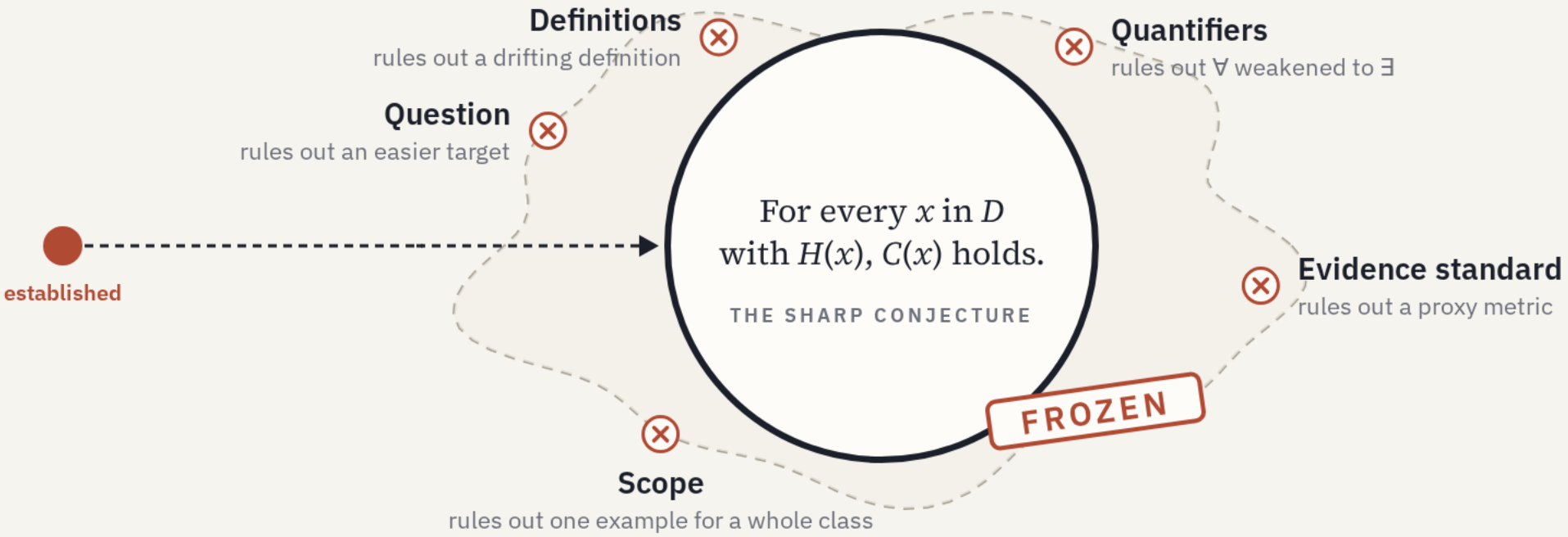
Organize: isolated thrusts, a gated core.



Each thrust is hard. How do we push through?

`formulate-problem`

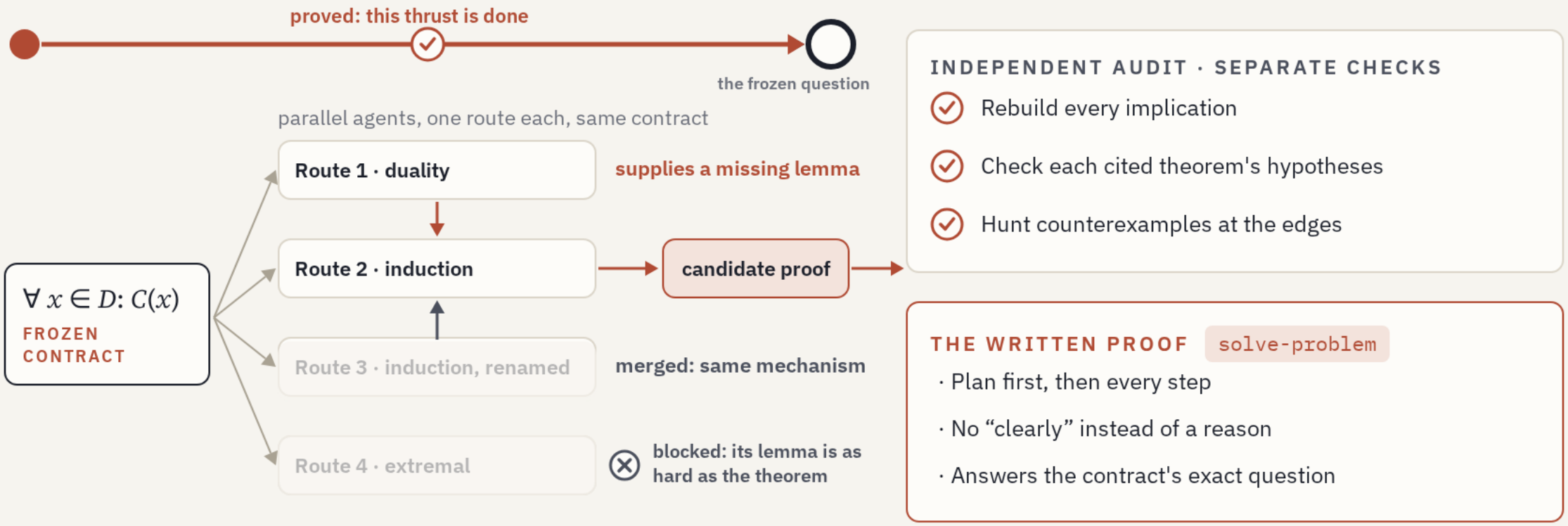
Formulate the steelman, then freeze it.



Now, how do we solve it?

`solve-problem`

Solve by many routes, then audit the proof.



Search and audit: `formulate-problem` · write-up: `solve-problem`

Solved. Do we understand it?

`teach-material`

`explain-canonically`

Learn the idea first, then every step.

The paper

$$\mathfrak{U}_{k+1} = \mathfrak{U}_k - \eta\, \mathfrak{D}(\mathfrak{U}_k),\ \mathfrak{D} := \partial\mathfrak{L}/\partial\mathfrak{U},\ \|\mathfrak{D}(\mathfrak{U}) - \mathfrak{D}(\mathfrak{V})\| \le \beta\|\mathfrak{U} - \mathfrak{V}\| \Longrightarrow \mathfrak{L}(\mathfrak{U}_k)\downarrow$$

Canonical notation

$$\theta_{k+1} = \theta_k - \eta\, \nabla L(\theta_k)$$

$\mathfrak{U} \to \theta \in \mathbb{R}^d$, the parameters · $\mathfrak{L} \to L : \mathbb{R}^d \to \mathbb{R}$, the loss · $\mathfrak{D} \to \nabla L$, its gradient · $\eta > 0$, the step size

The idea

Every step goes downhill; a smooth slope guarantees a definite drop.

plan: a drop per step → add up the drops

A toy example

$L(\theta) = \theta^2/2$ in one dimension: $\theta_{k+1} = (1 - \eta)\,\theta_k$, so $\theta_k \to 0$ exactly when $0 < \eta < 2$.



Theorems, checked

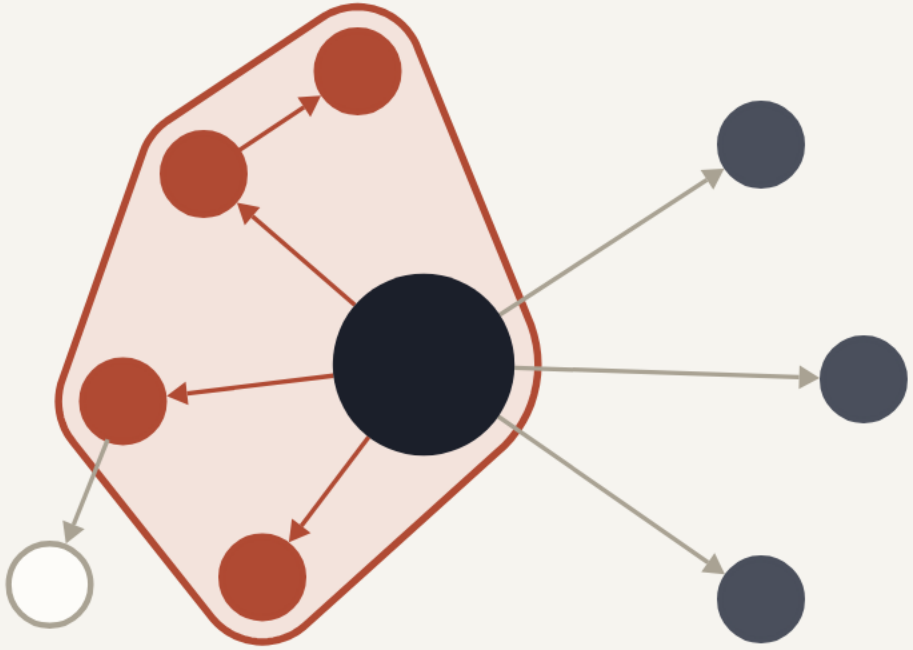
Descent lemma: if ∇L is β-Lipschitz, then $L(\theta - \eta\nabla L(\theta)) \le L(\theta) - \eta(1 - \beta\eta/2)\|\nabla L(\theta)\|^2$.

hypothesis ✓ β-Lipschitz, from the paper · with $\eta \le 1/\beta$: drop $\ge (\eta/2)\|\nabla L(\theta_k)\|^2$

Is it a paper?

review-paper

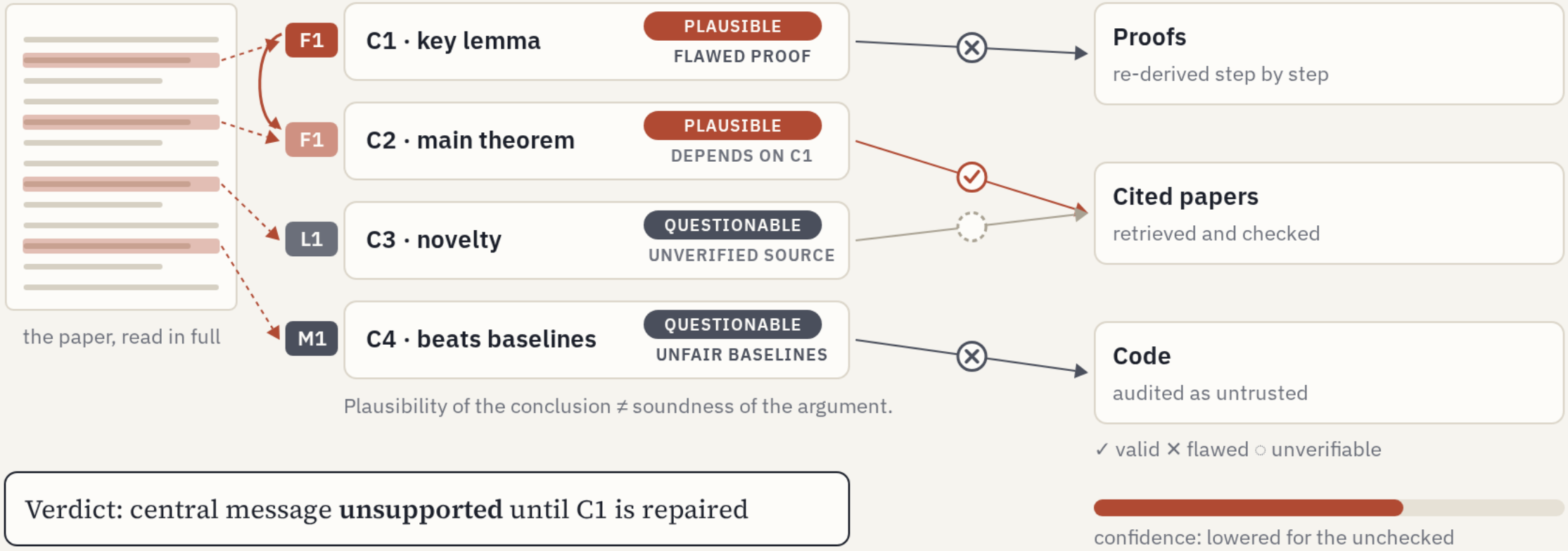
Audit every claim against its evidence.



THE ESTABLISHED CORE

`review-paper`

Audit every claim against its evidence.



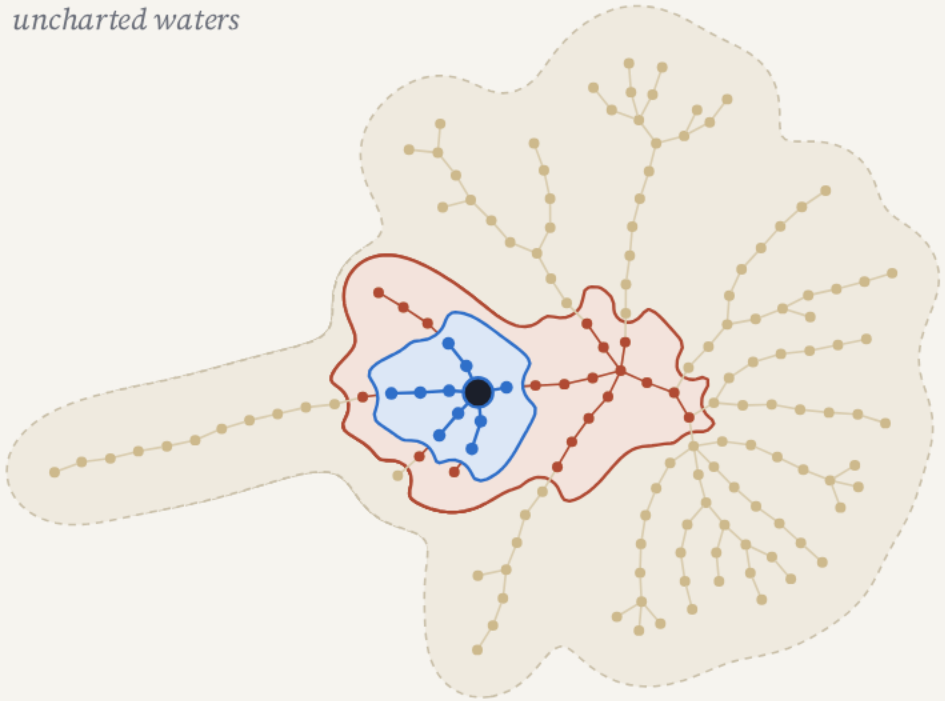
Verdict: central message **unsupported** until C1 is repaired

Putting it all together.

11 · PUTTING IT ALL TOGETHER

From the haystack to the core to the gem.

Open-ended question → (organize-research-project: formulate · solve · learn) → Research thrusts → (promotion gate) → Established core → (review-paper) → Carved-out paper



How to use it

Get

```
git clone \
  github.com/ajoudaki/frontier-research-skills
```

Install

```
ln -s $PWD/<skill> ~/.claude/skills/
# or ~/.codex/skills/ for Codex
```

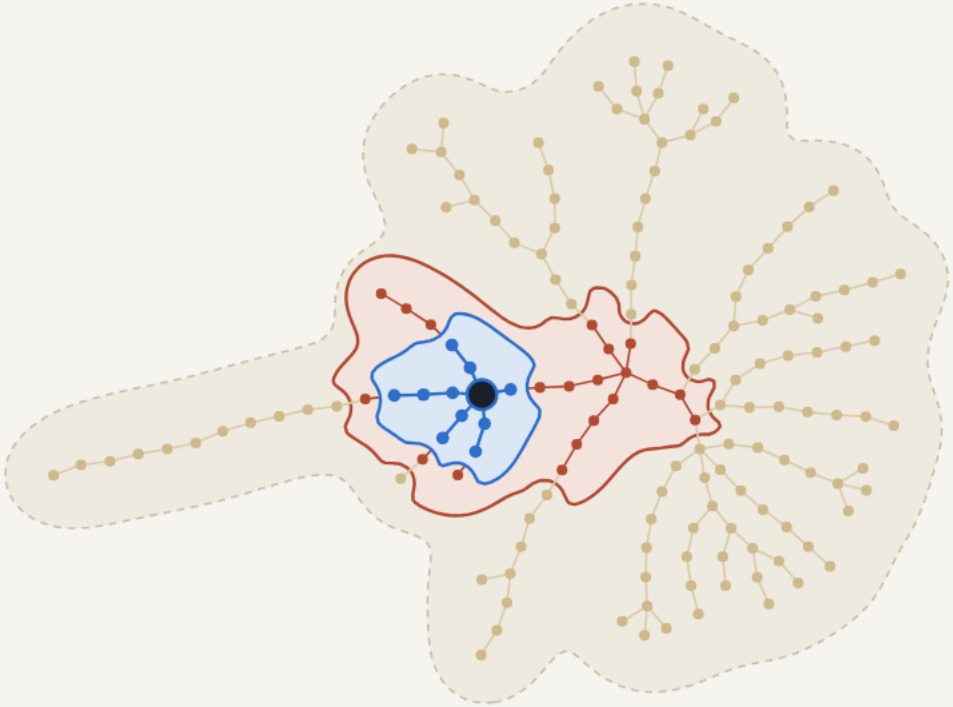
Or per project

```
name it in AGENTS.md or CLAUDE.md
```

The README has the details.

12 · CONCLUSION

From one question to a gem, and what's next.



open question → thrusts → core → gem

One goal: raise the signal-to-noise ratio.

WHAT'S NEXT

Beyond theory other domains

Evaluations does a change help a skill?

Live skills learn from your chats

THE BIG QUESTION

Can agents learn research taste?

github.com/ajoudaki/frontier-research-skills

Thank you.